

Andi Zhang

PH.D. CANDIDATE IN COMPUTER SCIENCE

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Education

University of Cambridge

Cambridge, United Kingdom

PH.D. IN COMPUTER SCIENCE

Oct. 2019 - Jul. 2023 (expected)

- Supervised by Dr. Damon Wischik.
- Ph.D project is about a Quasi-Bayesian approach to uncertainty in deep learning

MAST. IN MATHEMATICAL STATISTICS

Oct. 2018 - Jun. 2019

- Mathematical Tripos Part iii. Relevant modules:
Statistical Learning in Practise, Convex Optimization, Astrostatistics
Bayesian Modelling and Computation, Modern Statistical Methods

M.PHIL. IN ADVANCED COMPUTER SCIENCE

Oct. 2017 - Jun. 2018

- Relevant modules:
Machine Learning for Natural Language Processing, Algebraic Path
Deep Learning for Natural Language Processing, Advanced Functional Programming
Probabilistic Machine Learning

University of Manchester

Manchester, United Kingdom

B.SC. (HONS) IN COMPUTER SCIENCE AND MATHEMATICS

Sept. 2014 - Jun. 2017

- Relevant modules:
Machine Learning, Quantum Computing, Compilers
Convex Optimization, Probability, Statistical Methods
Functional Analysis, Topology, Number Theory

Experience

Noah's Ark Lab, Huawei

Shenzhen / Beijing, China

RESEARCH INTERN | MENTOR: DR. YITONG SUN

Sept. 2020 - Sept. 2021

- Worked on deep learning uncertainty, 3 papers are published during the internship.

Trinity College, University of Cambridge

Cambridge, United Kingdom

COURSE SUPERVISOR (TEACHING ASSISTANT)

Oct. 2019 - Jun. 2021

- Supervised IA Algorithms (2020 Lent), IA Machine Learning (2020 Lent)
and IA Introduction to Probability (2020 Easter)

MRC Biostatistics Unit, University of Cambridge

Cambridge, United Kingdom

RESEARCH ASSISTANT | SUPERVISOR: DR. SOFIA VILLAR AND DR. DAVID ROBERTSON

Jul. 2019 - Sept. 2019

- Built a simulation system for multi-armed bandit problem (MABP) in clinical trial.
- Presented a new test statistic for binary response MABP.

Tencent (WeChat AI)

Beijing, China

R & D INTERN | MENTOR: DR. JIE ZHOU

Jul. 2018 - Sept. 2018

- Concentrated on reinforcement learning and evolution strategy.

Amazon (Alexa)

Cambridge, United Kingdom

APPLIED SCIENTIST INTERN | MENTOR: DR. EMILIO MONTI

Jun. 2017 - Sept. 2017

- Concentrated on Neural Machine Translation (Sequence to sequence model).

Robotics Institute, Carnegie Mellon University

Pittsburgh, United States

STUDENT INTERN | SUPERVISOR: DR. STELIAN COROS

Jun. 2016 - Sept. 2016

- Assisted in implementing general stance controller for the 3D printable robotic creatures.
- Helped reproduce the prototype animation system.

Projects

A Quasi-Bayesian approach to uncertainty in deep learning

Cambridge, United Kingdom

PH.D. PROJECT | SUPERVISOR: DR. DAMON WISCHIK

Dec. 2019 - Now

- Presented a Quasi-Bayesian approach to capture epistemic uncertainty for deep learning.
- Rethought the relationship between uncertainty and calibration of the classification problem and present a new evaluation method for neural network calibration.

Visualizing and Clustering Data Centre Transactions

Cambridge, United Kingdom

M.PHIL. PROJECT | SUPERVISOR: DR. DAMON WISCHIK

Jan. 2018 - Jun. 2018

- Introduced several unsupervised deep learning models for business transactions.
- Set up three challenges in application level and solve them in a simulated application scenarios.
- The unsupervised deep learning models achieve a satisfying result in both visualization and clustering.

Add Uncertainty to Convolutional Neural Network by Gaussian Process

Cambridge, United Kingdom

COURSE PROJECT | SUPERVISOR: DR. DAMON WISCHIK

Jan. 2018 - Feb. 2018

- Coursework for LE49 Probabilistic Machine Learning.
- Proposed a GP classification model using CNN as its kernel implicitly.
- Claimed a “punitive” metric based on epistemic uncertainty, in which the proposed model performs far better.

Deep Reinforcement Learning on Video Games

Manchester, United Kingdom

B.SC. PROJECT | SUPERVISOR: DR. JONATHAN SHAPIRO

Jul. 2016 - Mar. 2017

- Implemented agents on OpenAI Gym and Malmo (Minecraft Agent Platform).
- Compared the effect of Deep Q Learning and Deep Recurrent Q Learning on different environments.

Stendhal Game

Manchester, United Kingdom

COURSEWORK FOR SOFTWARE ENGINEERING | SUPERVISOR: DR. SUZANNE EMBURY

Jan. 2016 - May. 2016

- Added some new items, NPCs and quests to the open source online game Stendhal.
- Designed and implemented a new efficiency quest mechanism for game designer.
- Refactored the existing quest code to suit the new quest mechanism.
- Won the Stendhal Quick Quest Challenge held in Software Engineering course (1st place).

Publications

On the Out-of-distribution Generalization of Probabilistic Image Modelling

Mingtian Zhang*, Andi Zhang*, Steven McDonagh

In Neural Information Processing Systems annual meeting (NeurIPS), 2021

Measuring Uncertainty in Signal Fingerprinting with Gaussian Processes Going Deep

Ran Guan, Andi Zhang, Mengchao Li, Yongliang Wang

In International Conference on Indoor Positioning and Indoor Navigation (IPIN), 2021

Towards dynamic and scalable active learning with neural architecture adaption for object detection

Fuhui Tang, Dafeng Wei, Chenhan Jiang, Hang Xu, Andi Zhang, Wei Zhang, Hongtao Lu, Chunjing Xu

In The British Machine Vision Conference (BMVC), 2021

Solipsistic Reinforcement Learning

Mingtian Zhang*, Peter Noel Hayes*, Tim Z. Xiao, Andi Zhang, David Barber

In Self-Supervision for Reinforcement Learning Workshop - International Conference on Learning Representations (ICLR), 2021

Honors & Awards

The Williams/Kilburn Medal for outstanding final year student

Manchester, United Kingdom

School of Computer Science, University of Manchester

Jul. 2017

The medal is awarded on an occasional basis to students of exceptional distinction whose performance throughout their undergraduate course has been of outstanding merit.

University awards for outstanding academic achievement

University of Manchester

The University will consider the top 0.5% of undergraduates (nominated by each School) for this award.